

[HLD] PRAJNA — High-Level Design Document

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- **Status:** Pending Review

Glossary

Term	Full Form	What It Means
HLD	High-Level Design	Architecture-level design document covering what and why
LLD	Low-Level Design	Detailed design document covering how — APIs, schemas, logic
CDK	AWS Cloud Development Kit	Framework to define AWS infrastructure using TypeScript code
PWA	Progressive Web App	A web app that works like a native mobile app — installable, offline-capable, push notifications
IaC	Infrastructure as Code	Managing cloud resources through code instead of manual setup
RBAC	Role-Based Access Control	Restricting access based on user roles (Faculty, HoD, Director, etc.)
JWT	JSON Web Token	A secure token used to authenticate API requests
SSO	Single Sign-On	Login once, access multiple systems
CI/CD	Continuous Integration / Continuous Deployment	Automated build, test, and deploy pipeline

Term	Full Form	What It Means
APAR	Annual Performance Appraisal Report	Yearly faculty performance review
NAAC	National Assessment and Accreditation Council	Body that accredits universities in India
NBA	National Board of Accreditation	Accredits technical programs in India
NIRF	National Institutional Ranking Framework	India's university ranking system
CO-PO	Course Outcome - Programme Outcome	Mapping of what a course teaches to overall program goals
UGC CAS	University Grants Commission - Career Advancement Scheme	Rules for faculty promotions in Indian universities
FDP	Faculty Development Programme	Training programs for faculty skill development
MOOC	Massive Open Online Course	Online courses (e.g., NPTEL, Coursera, Udemy)
IQAC	Internal Quality Assurance Cell	University body responsible for quality and accreditation readiness
DOI	Digital Object Identifier	Unique ID for academic publications
SES	Simple Email Service	AWS service for sending emails
SNS	Simple Notification Service	AWS service for sending SMS and push notifications
SCP	Service Control Policy	AWS Organization policy to restrict what services accounts can use

1. Introduction

Purpose

This document defines the high-level architecture and design of PRAJNA — the Professional AI Companion for GITAM Faculty (Phase 1), with expansion to

students in Phase 2. It serves as the architectural blueprint for the Super-30 student development team.

Scope

- Phase 1: Faculty companion across 3 campuses (Bengaluru, Vizag, Hyderabad)
- Development period: May–July 2026 (3 months)
- 30 student developers, each owning one module

How This Document Works

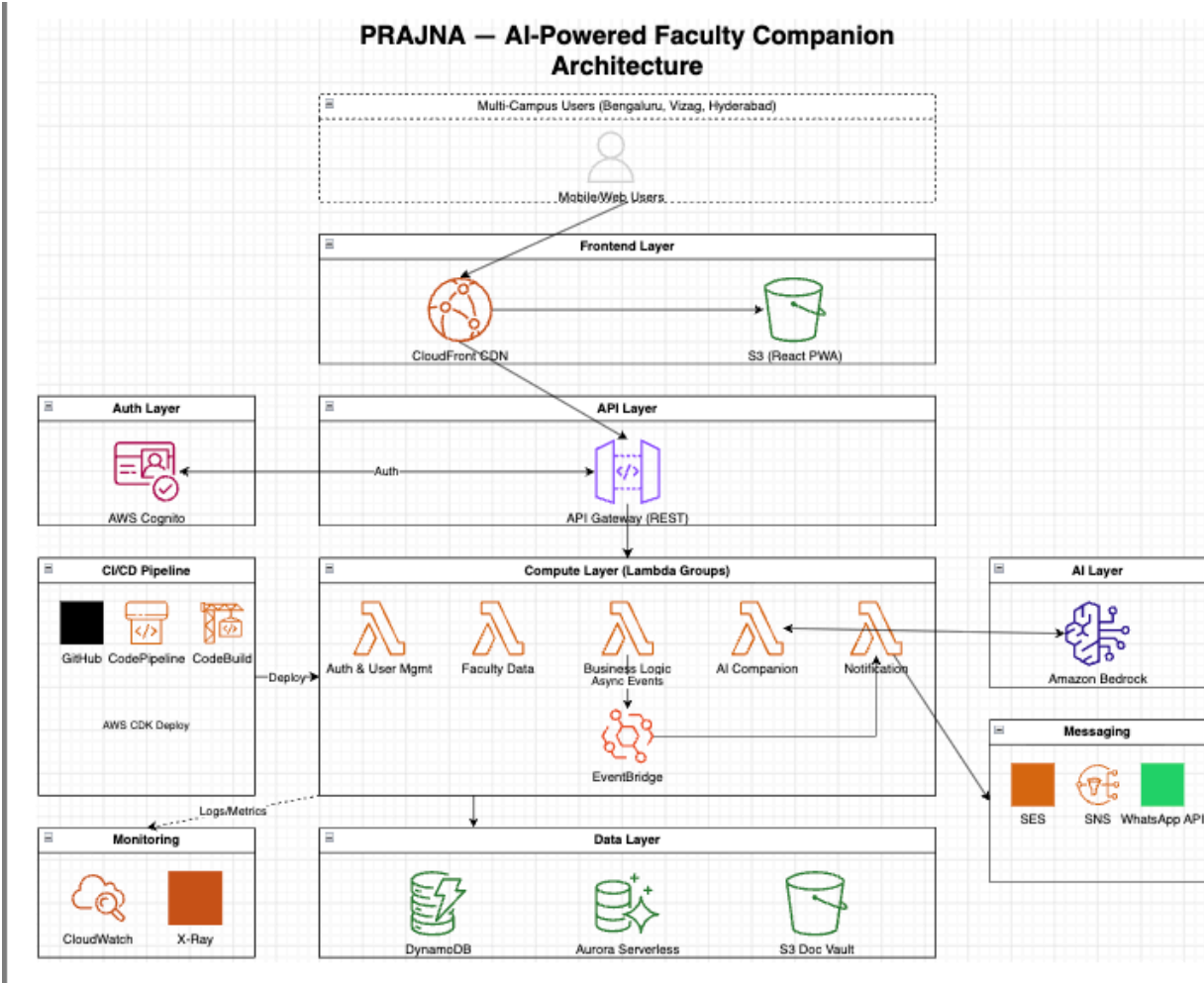
- **HLD** — architecture decisions, component interactions, and module definitions. Owned by the architect.
 - **LLD (per module)** — each student creates one using the . Covers detailed design, API contracts, data models, technology choices, and testing.
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2. Architecture Overview

Architecture Style

Serverless-first modular architecture on AWS. Each module is independently deployable through its own CDK stack and communicates via API Gateway, EventBridge, and shared data stores.

High-Level Component Map



Key Architecture Decisions

Decision	Choice	Why
Architecture style	Serverless modular	Zero server management, pay-per-use, each module deploys independently
Language	TypeScript only	Single language across frontend, backend, and CDK infra
IaC	AWS CDK	TypeScript-native, reusable constructs, one language everywhere
Database (NoSQL)	DynamoDB	Serverless, auto-scaling, pay-per-request
Database (SQL)	Aurora Serverless v2	Relational queries for reports, NAAC/NBA compliance data
AI	Amazon Bedrock	Managed LLM access, no API key

Decision	Choice	Why
		management, stays within AWS
Auth	Cognito	Serverless, integrates natively with API Gateway and Lambda
Frontend hosting	CloudFront + S3	Serverless, global CDN, zero servers
Event-driven communication	EventBridge	Decouples modules, async processing for notifications/workflows
File storage	S3	Serverless, lifecycle policies, pre-signed URLs for secure access

3. Architecture Constraints

These apply to all modules and are documented in detail in the [LLD Template]

- **TypeScript only** — all code and libraries
- **AWS Serverless only** — Lambda, API Gateway, DynamoDB, Aurora Serverless v2 (SQL), S3, Cognito, SQS, SNS, EventBridge, Step Functions, Amazon Bedrock (AI) etc.
- **Each module owns its own CDK stack**
- **Any other exception requires written approval from the architect**

4. Component Architecture

4.1 Frontend

- React + TypeScript PWA
- Hosted on S3 + CloudFront
- Mobile-first, web- and mobile-compatible
- Communicates with backend via API Gateway

4.2 API Layer

- API Gateway (REST)
- Lambda authorizers for auth

- Each module exposes its own endpoints
- Request validation at the gateway level

4.3 Auth & Authorization

- AWS Cognito for user management
- Role-based access: Faculty, HoD, Director (school), Pro Vice-Chancellor (campus), IQAC, Admin (all campuses)
- Multi-campus data isolation
- JWT tokens for API auth

4.4 Data Layer

- DynamoDB for most module data (profiles, achievements, FDPs, etc.)
- Aurora Serverless v2 for complex relational queries (reports, analytics, cross-module joins)
- S3 for document/certificate storage

4.5 AI Companion

- Amazon Bedrock for LLM capabilities
- Companion chat, morning briefings, end-of-day summaries, and career guidance
- Rule-based fallback if AI is unavailable
- AI conversations are private to the faculty member
- **Multi-channel access** — designed to sync data across all communication channels:
 - **Phase 1:** In-app chat (primary UI) — faculty start here
 - **Future:** WhatsApp, Email, Microsoft Teams, and other platforms used by GITAM (expanded based on faculty feedback on usefulness)
 - All channels sync to one conversation history

- **Dynamic To-Do Engine** — PRAJNA generates a personalized daily to-do list for each faculty member based on:
 - Upcoming classes and attendance to be marked
 - Assessments to be graded
 - Exam schedules and question paper preparation deadlines
 - Pending approvals and submissions
 - Research and FDP deadlines
 - APAR and compliance tasks
- The to-do list updates dynamically as tasks are completed, reprioritizing the remaining items and surfacing the next most important action
- Faculty can view, complete, and interact with their to-do list from any channel

4.6 Event & Notification Layer

- EventBridge for inter-module events (e.g., "publication approved" triggers score recalculation, notification, and leaderboard update)
- SES for email, SNS for SMS
- WhatsApp via Meta Business API (Lambda integration)
- In-app notifications via DynamoDB + WebSocket/polling

4.7 Business Logic Layer

- **Approval Workflow Engine** — multi-level approval flows (Faculty → HoD → Director), auto-escalation on missed deadlines, configurable timeout windows, auto-approve rules
- **PRAJNA Score Engine** — real-time composite score (Teaching 25%, Research 30%, FDP 20%, Achievements 10%, Admin 10%, Profile Completeness 5%), tier assignment (Bronze → PRAJNA Fellow), recalculated on every data change via EventBridge
- **Leaderboard** — department, school, and cross-campus rankings, monthly reset + all-time board, designed to inspire, not shame

- **APAR & Appraisal** — annual appraisal lifecycle, self-assessment with AI-suggested inputs, sequential HoD → Director review, API score calculation (UGC CAS)

4.8 Report Engine

- NAAC criterion-wise reports (Criteria I–VII) auto-populated from live data
- NBA OBE reports with CO-PO mapping
- NIRF teaching, research, and graduation outcomes data
- Custom analytics exports
- PDF and Excel generation via Lambda
- Continuous inspection readiness score (0–100%) per department, school, and campus

4.9 Data Migration

- Excel import templates with validation
- Bulk import scripts for historical data
- Data verification dashboards for HoD sign-off
- Records with missing mandatory fields are flagged, not auto-imported
- Historical data (pre-2020) is imported as 'Archived' — visible but not editable
- Original files are preserved in S3 for 7 years

4.10 Monitoring & Observability

- CloudWatch for logs, metrics, and alarms
- X-Ray for distributed tracing across Lambdas
- CloudWatch dashboards for system health
- AI response quality monitoring and prompt evaluation

5. Data Flow

5.1 Faculty Login & Role-Based Routing

1. Faculty opens React PWA
2. Cognito handles login (OTP / SSO)
3. Cognito returns JWT with role + campus claims
4. React reads JWT and routes to role-specific dashboard:
 - Faculty → Faculty Dashboard
 - HoD → HoD Dashboard
 - Director → School Dashboard
 - Pro Vice-Chancellor → Campus Dashboard
 - Admin → All Campuses Dashboard
5. All subsequent API calls include JWT
6. API Gateway → Cognito Authorizer validates token
7. Lambda reads role + campus from token → filters data accordingly

5.2 Faculty Adds a Publication

1. Faculty enters DOI in React UI
2. API Gateway → Lambda (Research Module)
3. Lambda validates and checks for duplicate in DynamoDB
4. Auto-fetches publication details from CrossRef/Scopus API
5. Faculty confirms details, uploads PDF proof
6. Saved to DynamoDB with status: pending
7. EventBridge → Notification Lambda → notifies HoD
8. HoD reviews and approves:
 - If SCI/Scopus → auto-escalates to Director (school level)
 - If Conference → HoD approval is final
9. On final approval, EventBridge triggers:

- Score Engine Lambda → recalculates PRAJNA Score
- Leaderboard Lambda → updates rankings
- AI Companion Lambda → sends celebration message
- To-Do Engine Lambda → removes from pending, surfaces next action

5.3 Approval Workflow with Escalation

1. Faculty submits data (publication, FDP cert, award, etc.)
2. Saved to DynamoDB with status: pending_approval
3. EventBridge → Notification Lambda → notifies Level 1 approver (HoD)
4. Escalation timeline:
 - Day 0 → HoD notified
 - Day 3 → Reminder sent to HoD
 - Day 7 → If no action: auto-escalate to Director, OR auto-approve with audit log (for FDP certs)
 - Day 10 → If still no action: escalate to Pro Vice-Chancellor, flag on Admin dashboard
5. On approval/rejection at any level:
 - EventBridge → Score Engine (if approved)
 - EventBridge → Notification Lambda → notifies faculty of outcome
 - EventBridge → AI Companion → celebration or guidance message

5.4 Morning Briefing

1. CloudWatch Scheduled Event fires at 8 AM daily
2. Lambda (AI Companion) reads faculty profile, pending tasks, deadlines from DynamoDB
3. Reads dynamic to-do list
4. Generates personalized briefing via Bedrock
5. Saves briefing to DynamoDB

6. EventBridge → Notification Lambda → pushes to faculty (in-app)

5.5 Dynamic To-Do Generation

Triggers:

- Faculty logs in (start of day)
- Task completed (real-time reprioritization)
- New deadline approaching (scheduled check)
- New data submitted by/for faculty (EventBridge event)

Lambda (To-Do Engine) queries DynamoDB for:

- Today's classes and attendance status
- Pending assessments to grade
- Upcoming exam / question paper prep deadlines
- Pending approval submissions
- Research and FDP deadlines
- APAR and compliance tasks

Processing:

1. Ranks items by urgency and importance
2. Saves prioritized to-do list to DynamoDB
3. Pushes update to faculty via WebSocket / in-app notification

On task completion:

1. Faculty marks task done → Lambda removes it
2. Re-ranks remaining items → surfaces next most important action
3. AI Companion → "Great work! Next up: [next task]"

6. Multi-Campus & Multi-Tenancy

- Campus isolation via partition keys in DynamoDB / row-level filtering in Aurora
 - Role-based visibility:
 - Faculty → own data only
 - HoD → own department
 - Director → own school
 - IQAC → inspection readiness data across departments
 - Pro Vice-Chancellor → own campus
 - Admin → all campuses
 - Cognito user groups per campus + role
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7. Security

- Encryption at rest (AWS managed keys) and in transit (TLS 1.3)
 - Cognito MFA for admin roles
 - API Gateway request throttling
 - Lambda IAM roles with least privilege
 - AI companion conversations are private and not visible to anyone except the faculty member
 - Immutable audit trail for all data changes (DynamoDB streams)
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8. Deployment

- Each module = its own CDK stack
 - Environments: dev → staging → prod
 - CI/CD via AWS CodePipeline + CodeBuild
 - Each student deploys to their own dev account (AWS Organization)
 - Staging & prod on shared account managed by architect
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9. Module Breakdown — 30 Ownership Areas

Each module is assigned to one student. Every student creates an LLD using the [LLD Template] for their module.

Core Platform

#	Module	Description	Dependencies
1	CDK Foundation & Shared Constructs	Shared CDK constructs, environment config, tagging standards	None — other modules depend on this
2	CI/CD Pipeline	CodePipeline + CodeBuild setup for all module stacks	Module 1
3	Auth & User Management	Cognito setup, user pools, role groups, login/signup flows	Module 1
4	API Gateway & Middleware	Shared API Gateway, Lambda authorizers, request validation, CORS	Modules 1, 3
5	Database Layer	DynamoDB table designs, Aurora Serverless setup, access patterns	Module 1
6	File Storage & Document Vault	S3 buckets, pre-signed URL generation, virus scanning, lifecycle policies	Module 1

Faculty Data Modules

#	Module	Description	Dependencies
7	Personal & Professional Profile	Faculty profile CRUD, qualifications, ORCID/Scopus IDs, photo upload	Modules 3, 4, 5, 6
8	Course Deliverables & Teaching	Timetable, teaching load, lesson plans, attendance, CO-PO mapping, faculty advisor logs	Modules 3, 4, 5
9	Research & Innovation	Publications (DOI auto-fetch via CrossRef/Scopus API), duplicate	Modules 3, 4, 5, 6

#	Module	Description	Dependencies
		detection with side-by-side comparison, grants, patents, Ph.D. scholars	
10	Achievements & Recognition	Awards, invited talks, professional memberships, editorial roles	Modules 3, 4, 5
11	Faculty Development & Growth	FDPs, MOOCs, certifications, workshops, international visits	Modules 3, 4, 5, 6
12	Administrative & Lifecycle	Leave records, onboarding, exit, committee roles, exam duties	Modules 3, 4, 5

Business Logic

#	Module	Description	Dependencies
13	Approval Workflow Engine	Multi-level approval flows (Faculty → HoD → Director), auto-escalation, timeouts	Modules 4, 5, 19
14	PRAJNA Score Engine	Composite score calculation, tier assignment, real-time updates on data changes	Modules 5, 7–12
15	Leaderboard System	Department, school, cross-campus rankings, monthly reset, all-time board	Modules 5, 14
16	Notification Engine	Email (SES), SMS (SNS), WhatsApp (Meta API), in-app notifications	Modules 4, 5, 19
17	Report Generator	NAAC, NBA, NIRF reports, custom analytics, PDF/Excel export	Modules 5, 7–12
18	APAR & Appraisal Workflow	Annual appraisal flow, self-assessment, HoD → Director review, API score calculation	Modules 5, 13, 14

AI Companion

#	Module	Description	Dependencies
19	Event Bus & Integration Layer	EventBridge setup, event schemas, inter-module event routing	Module 1
20	AI Companion Chat & Fallback	Bedrock integration, conversational UI, context-aware responses from faculty profile, rule-based fallback when Bedrock is unavailable	Modules 4, 5, 7
21	Morning Briefing & End-of-Day	Scheduled Lambda, personalized daily agenda, sign-off summary, reflection capture	Modules 5, 16, 20
22	Career Coach & Opportunity Spotter	Gap analysis vs promotion criteria, conference/grant recommendations, workload balance monitoring, detects upcoming conferences/grants/FDPs relevant to faculty's research area	Modules 5, 14, 20
23	Dynamic To-Do Engine	Personalized daily to-do list based on classes, assessments, exams, question paper prep, approvals, deadlines. Dynamically reprioritizes as tasks are completed	Modules 5, 7–12, 20

Dashboards

#	Module	Description	Dependencies
24	Faculty Dashboard	Home screen, today's priorities, profile completeness, PRAJNA Score, personal timeline	Modules 7, 14, 15
25	HoD Dashboard	Department overview, approval queue, faculty needing attention, workload balance	Modules 7, 13, 14, 15
26	Director, Pro VC & IQAC Command Centre	Director: school-level view. Pro Vice-Chancellor: campus-level view. IQAC: inspection readiness score (0–100%), NAAC criterion-wise view. Shared: real-time	Modules 14, 15, 17

#	Module	Description	Dependencies
		faculty activity pulse, top performers, weekly auto-generated executive summary, cross-campus benchmarking	
27	Multi-Channel Integration	Sync PRAJNA across communication channels (in-app first, then WhatsApp, Email, Teams based on faculty feedback)	Modules 4, 16, 20

Cross-Cutting

#	Module	Description	Dependencies
28	Security & Audit Trail	DynamoDB streams for audit log, data change tracking, compliance reporting	Modules 4, 5
29	Monitoring & Observability	CloudWatch dashboards, X-Ray tracing, alarms, error tracking, AI response quality testing and prompt evaluation	Modules 1, 2
30	Data Migration Tools	Import scripts, Excel template validators, data verification dashboards	Modules 5, 7–12

Bonus Modules (For students looking for additional work)

#	Module	Description	Dependencies
B1	GITAM Portal & HR Integration	Bridge module to sync with existing GITAM Portal, HR leave system, and LMS (Moodle). GITAM Portal is master for shared data. Fallback to standalone if integration is delayed.	Modules 4, 5, 12
B2	PWA & Mobile Optimization	Service workers, offline storage, push notifications, mobile-specific UX optimizations for key actions (attendance, certificate upload)	Modules 4, 24

10. Module Assignment Tracker

#	Module	Assigned To	LLD Status	Dev Status
1	CDK Foundation & Shared Constructs			
2	CI/CD Pipeline			
3	Auth & User Management			
4	API Gateway & Middleware			
5	Database Layer			
6	File Storage & Document Vault			
7	Personal & Professional Profile			
8	Course Deliverables & Teaching			
9	Research & Innovation			
10	Achievements & Recognition			
11	Faculty Development & Growth			
12	Administrative & Lifecycle			
13	Approval Workflow Engine			
14	PRAJNA Score Engine			
15	Leaderboard System			
16	Notification Engine			

#	Module	Assigned To	LLD Status	Dev Status
17	Report Generator			
18	APAR & Appraisal Workflow			
19	Event Bus & Integration Layer			
20	AI Companion Chat & Fallback			
21	Morning Briefing & End-of-Day			
22	Career Coach & Opportunity Spotter			
23	Dynamic To-Do Engine			
24	Faculty Dashboard			
25	HoD Dashboard			
26	Director, Pro VC & IQAC Command Centre			
27	Multi-Channel Integration			
28	Security & Audit Trail			
29	Monitoring & Observability			
30	Data Migration Tools			
B1	GITAM Portal & HR Integration			
B2	PWA & Mobile Optimization			

- **LLD Status:** Not Started | In Progress | In Review | Approved

- **Dev Status:** Not Started | In Progress | Testing | Done

11. Timeline

Phase	Period	What Happens
LLD Phase	Week 1–2	All 30 students write their LLDs. Reviewed and approved by mentors.
Foundation Build	Week 3–4	Core platform modules (1–6) are built first. Others continue refining LLDs and begin scaffolding CDK stacks.
Module Build	Week 5–8	All modules are in active development. Weekly integration checkpoints.
Integration & Testing	Week 9–10	Modules connect. End-to-end testing. Bug fixes.
UAT & Polish	Week 11–12	UAT with faculty and HoDs. Final fixes. Go-live preparation.--

12. References

- PRAJNA v3 Requirements Document
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PRAJNA — प्रज्ञा / Super-30 — Building industry-ready engineers through real-world experience